

Enzo Stefanelli Negjuni 9,3/10
Set 1

13689372
QFT I

1.4) Consider an infinitesimal Lorentz transformation

1,9/2

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu, \quad \omega^\mu_\nu \ll 1$$

In order that $x^\mu x_\mu = x'^\mu x'_\mu = g_{\alpha\beta} \Lambda^\alpha_\mu x^\mu \Lambda^\beta_\nu x^\nu$

$$\Rightarrow g_{\mu\nu} = \Lambda^\alpha_\mu \Lambda^\beta_\nu g_{\alpha\beta}$$

$\therefore$

$$g_{\mu\nu} = g_{\alpha\beta} (\delta^\alpha_\mu + \omega^\alpha_\mu) (\delta^\beta_\nu + \omega^\beta_\nu)$$

Keeping the leading-order terms

$$g_{\mu\nu} = g_{\mu\nu} + \omega_{\mu\nu} + \omega_{\nu\mu} \Rightarrow \omega_{\mu\nu} = -\omega_{\nu\mu}$$

$\downarrow$
Antisymmetric

The elements of the group are, therefore,

$$\Lambda \equiv \exp\left(\frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu}\right)$$

and for an infinitesimal transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \omega^\mu_\nu x^\nu$$

$$\phi(x^\mu) \rightarrow \phi'(x'^\mu) = \phi(x^\mu + \omega^\mu_\nu x^\nu) = \phi(x^\mu) + \omega^\mu_\nu x^\nu \partial_\mu \phi(x^\mu)$$

in the expansion there should be a prime in ϕ , then, you should expand again.

$$\Rightarrow \delta\phi = \omega^\mu_\nu x^\nu \partial_\mu \phi(x^\mu) \quad \textcircled{I}$$

Moreover $\phi'(x') = \Lambda \phi(x)$

$$\phi'(x') = \Lambda \phi(x) = \phi(x) + \frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu} \phi(x)$$

$$\Rightarrow \delta\phi = \frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu} \phi(x) \quad \textcircled{II}$$

$\left(1 + \frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu}\right)$

Therefore: $\phi'(x') = \Lambda \phi(x)$

but: $\phi'(x') = \phi(x) \Rightarrow \Lambda = 1$

From equations ① and ② $\phi(x^\mu) \equiv \phi$ scalar Product are invariants

$$\frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu} \phi = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi = \omega_{\mu\nu} x^\nu \partial^\mu \phi$$

Since $\omega_{\mu\nu}$ is antisymmetric

$$\Rightarrow \frac{1}{2} i \omega_{\mu\nu} \mathcal{J}^{\mu\nu} \phi = \frac{1}{2} \omega_{\mu\nu} (x^\nu \partial^\mu \phi - x^\mu \partial^\nu \phi)$$

So then

$$\mathcal{J}^{\mu\nu} = -i (x^\mu \partial^\nu - x^\nu \partial^\mu) \quad (\text{Antisymmetric})$$

there is no problem to define with an opposite sign, you just have to be careful with $\Lambda = \exp(\pm i \omega \frac{\mathcal{J}}{2})$ and whether you're considering an active vs passive transformation

Now it's possible to evaluate

$$i[\mathcal{J}^{\alpha\beta}, \mathcal{J}^{\rho\sigma}] = -i([x^\alpha \partial^\beta, x^\rho \partial^\sigma] - [x^\alpha \partial^\beta, x^\sigma \partial^\rho] - [x^\beta \partial^\alpha, x^\rho \partial^\sigma] + [x^\beta \partial^\alpha, x^\sigma \partial^\rho])$$

Knowing that $[x^\mu, x^\nu] = [\partial^\mu, \partial^\nu] = 0$, the non-vanishing terms are

$$= -i(x^\alpha [\partial^\beta, x^\rho] \partial^\sigma + x^\rho [x^\alpha, \partial^\sigma] \partial^\beta - x^\alpha [\partial^\beta, x^\sigma] \partial^\rho - x^\sigma [x^\alpha, \partial^\rho] \partial^\beta - x^\beta [\partial^\alpha, x^\rho] \partial^\sigma - x^\rho [x^\beta, \partial^\sigma] \partial^\alpha + x^\beta [\partial^\alpha, x^\sigma] \partial^\rho + x^\sigma [x^\beta, \partial^\rho] \partial^\alpha)$$

Using that $[\partial^\mu, x^\nu] = g^{\mu\nu}$ would be nice to have proof

$$= -i(g^{\beta\rho} x^\alpha \partial^\sigma - g^{\alpha\sigma} x^\rho \partial^\beta - g^{\beta\sigma} x^\alpha \partial^\rho + g^{\alpha\rho} x^\sigma \partial^\beta - g^{\alpha\rho} x^\beta \partial^\sigma + g^{\beta\sigma} x^\rho \partial^\alpha + g^{\alpha\sigma} x^\beta \partial^\rho - g^{\beta\rho} x^\sigma \partial^\alpha)$$

$$= g^{\beta\rho} i(x^\sigma \partial^\alpha - x^\alpha \partial^\sigma) + g^{\alpha\sigma} i(x^\rho \partial^\beta - x^\beta \partial^\rho) + g^{\beta\sigma} i(x^\alpha \partial^\rho - x^\rho \partial^\alpha) + g^{\alpha\rho} i(x^\beta \partial^\sigma - x^\sigma \partial^\beta)$$

$$= g^{\beta\rho} \mathcal{J}^{\sigma\alpha} + g^{\alpha\sigma} \mathcal{J}^{\rho\beta} + g^{\beta\sigma} \mathcal{J}^{\alpha\rho} + g^{\alpha\rho} \mathcal{J}^{\beta\sigma} \quad \checkmark$$

b) The generators of the Lorentz group may be separated into:

$$J_i = \frac{1}{2} \epsilon_{ijk} J^{jk} \quad (\text{Rotations})$$

$$K_i = J^{0i} \quad (\text{Boosts})$$

careful with spatial index placements, specially if you use $g = + - - -$

Where $i, j, k = 1, 2 \text{ or } 3$, and ϵ_{ijk} is the Levi-Civita tensor ($\epsilon_{123} = +1$). With these definitions the Lorentz Group is divided among the relations

$$\textcircled{1} [J_i, J_j] = \frac{1}{4} \epsilon_{ijk} \epsilon_{lmn}^{i \cdot j} [J^{kl}, J^{mn}]$$

$$= \frac{-i}{4} \epsilon_{ijk} \epsilon_{lmn} (\underbrace{\delta^{kl}}_{\textcircled{A}} J^{mj} + \underbrace{\delta^{jm}}_{\textcircled{B}} J^{kl} + \underbrace{\delta^{kn}}_{\textcircled{C}} J^{lm} + \underbrace{\delta^{ln}}_{\textcircled{D}} J^{km})$$

$$\textcircled{A}: \epsilon_{ijk} \epsilon_{lmn} \delta^{kl} J^{mj} = -\epsilon_{ijk} \epsilon_{lmn} \delta^{km} J^{nj}$$

$$= -\epsilon_{ijk} \epsilon_{lmn} J^{mj} = \epsilon_{ijk} \epsilon_{lmn} J^{mj} \quad (\text{Antisymmetric})$$

$$= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) J^{mj} = \delta_{il} J^{jj} - J^{il}$$

$$\textcircled{B}: \epsilon_{ijk} \epsilon_{lmn} \delta^{jm} J^{kl} = -\epsilon_{ijk} \epsilon_{lmn} J^{kl}$$

$$= (\delta_{il} \delta_{km} - \delta_{im} \delta_{kl}) J^{kl} = \delta_{il} J^{kk} - J^{il}$$

$$\textcircled{C}: \epsilon_{ijk} \epsilon_{lmn} \delta^{kn} J^{lm} = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) J^{lm} = -J^{il}$$

$$\textcircled{D}: \epsilon_{ijk} \epsilon_{lmn} \delta^{ln} J^{km} = (\delta_{il} \delta_{km} - \delta_{im} \delta_{kl}) J^{km} = -J^{il}$$

$$\therefore [J_i, J_j] = +\frac{i}{4} 4 J^{il} = i J^{il} = i \epsilon_{ilk} J_k$$

$$* J_i = \frac{1}{2} \epsilon_{ijk} J^{jk}$$

$$\Rightarrow J^{jk} = \epsilon_{ijk} J_i$$

$$\textcircled{II} [K_i, K_j] = [\sigma^{0i}, \sigma^{0j}] = \overset{\text{(Diagonal)}}{\left(\delta_{ij}^0 \sigma^{j0} + \delta_{ij}^0 \sigma^{0i} + \delta_{ij}^0 \sigma^{j0} + \delta_{ij}^0 \sigma^{0i} \right)} \\ = -i \sigma^{ij} = -i \epsilon_{ijk} \sigma_k$$

Note that from $\textcircled{II}$ the generators are entangled;
to get a simpler view of the irreducible representations,
we define

$$\sigma_i^{\pm} = \frac{1}{2} (\sigma_i \pm i K_i)$$

and evaluate

$$\textcircled{III} [\sigma_i, K_j] = \frac{1}{2} \epsilon_{ijk} [\sigma_i^{\pm}, \sigma_j^{\pm}] = \\ = \frac{-i}{2} \epsilon_{ijk} \left(\delta_{ij}^0 \sigma^{jl} + \delta_{ij}^0 \sigma^{0k} + \delta_{ij}^0 \sigma^{j0} + \delta_{ij}^0 \sigma^{kl} \right) \\ = \frac{i}{2} \epsilon_{ijk} \left(\delta^{jl} \sigma^{0k} + \delta^{kl} \sigma^{j0} \right) \overset{\sigma - \sigma^{0k} = -K_k}{=} = \frac{i}{2} (\epsilon_{ijk} K_k - \epsilon_{ijl} K_l) \\ \quad \quad \quad - \epsilon_{ijl} K_l$$

Calling the contract index as k

$$= \frac{i}{2} (\epsilon_{ijk} K_k + \epsilon_{ikl} K_k) = i \epsilon_{ijk} K_k \quad \text{OK}$$

Now using $\textcircled{I}$, $\textcircled{II}$ and $\textcircled{III}$; the new set of generator
give us

$$[\sigma_i^+, \sigma_j^-] = \frac{1}{4} [\sigma_i + i K_i, \sigma_j - i K_j] = \\ = \frac{1}{4} ([\sigma_i, \sigma_j] - i [\sigma_i, K_j] + i [K_i, \sigma_j] + [K_i, K_j]) \\ = \frac{1}{4} \epsilon_{ijk} (i \sigma_k + K_k - K_k - i \sigma_k) = 0$$

permutation
of Lorentz

$$\begin{aligned}
 [J_i^+, J_j^+] &= \frac{1}{4} [J_i + iK_i, J_j + iK_j] \\
 &= \frac{1}{4} ([J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] - [K_i, K_j]) \\
 &= \frac{i\epsilon_{ijk}}{4} (J_k + iK_k + iK_k + J_k) \\
 &= i\epsilon_{ijk} \left(\frac{J_k + iK_k}{2} \right) = i\epsilon_{ijk} J_k^+
 \end{aligned}$$

$$\begin{aligned}
 [J_i^-, J_j^-] &= \frac{1}{4} ([J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] - [K_i, K_j]) \\
 &= \frac{i\epsilon_{ijk}}{4} (J_k - iK_k - iK_k + J_k) \\
 &= i\epsilon_{ijk} \left(\frac{J_k - iK_k}{2} \right) = i\epsilon_{ijk} J_k^-
 \end{aligned}$$

We obtained now two set of generator, not entangled, that each one individually satisfy the $SU(2)$ algebra, based on this the irreducible representations of the Lorentz Group $SO(1,3)$ can be labeled in terms of the well known representations of $SU(2)$ by the pair (m, n) where $m, n = 0, \pm 1/2, \pm 3/2, \dots$. In other words we verify that $so(1,3) = su(2) \oplus su(2)$; some examples are

$su(2) \oplus su(2)$	$so(1,3)$
$(0, 0)$	0
$(0, \pm 1/2)$	$\pm 1/2$
$(\pm 1/2, 0)$	$\pm 1/2$
$(\pm 1/2, \pm 1/2)$	$1 \oplus 0$
$(1, 0)$	1
$(1, 1)$	$2 \oplus 1 \oplus 0$

OK!

2 A) To obtain the equations of motion we use
1, 6/2 the Euler-Lagrange equation

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0$$

In this case

$$\mathcal{L} = -\frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} = -\frac{1}{4} (\partial_\rho A_\sigma - \partial_\sigma A_\rho) (\partial^\rho A^\sigma - \partial^\sigma A^\rho)$$

$$\therefore \frac{\partial \mathcal{L}}{\partial A_\nu} = 0$$

$$g^{\alpha\rho} g^{\beta\sigma} (\partial_\alpha A_\beta - \partial_\beta A_\alpha)$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \stackrel{\text{chain rule}}{=} -\frac{1}{4} \left[(\delta^\mu_\rho \delta^\nu_\sigma - \delta^\mu_\sigma \delta^\nu_\rho) (\partial^\rho A^\sigma - \partial^\sigma A^\rho) + (\partial_\rho A_\sigma - \partial_\sigma A_\rho) g^{\alpha\rho} g^{\beta\sigma} (\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha) \right]$$

$$= -\frac{1}{4} [(\partial^\mu A^\nu - \partial^\nu A^\mu) - (\partial^\nu A^\mu - \partial^\mu A^\nu) + (\partial^\mu A^\nu - \partial^\nu A^\mu) - (\partial^\nu A^\mu - \partial^\mu A^\nu)]$$

$$= -\frac{1}{4} [2F^{\mu\nu} - 2F^{\nu\mu}] = -\frac{1}{4} [2F^{\mu\nu} + 2F^{\mu\nu}] = -F^{\mu\nu}$$

$\therefore$ The equations of motion are

$$\partial_\mu F^{\mu\nu} = 0$$

$$3) x^\mu \rightarrow x'^\mu = x^\mu + a^\mu \quad \text{Taylor}$$

$$\phi^\sigma(x) \rightarrow \phi^\sigma(x') = \phi^\sigma(x+a) = \phi^\sigma(x) + \frac{\partial \phi^\sigma}{\partial x^\mu} a^\mu \therefore \delta \phi^\sigma = a^\mu \partial_\mu \phi^\sigma$$

$$\text{again: } \delta \phi = \phi'(x) - \phi(x) = -a^\mu \partial_\mu \phi$$

By the Noether Theorem

$$J^\nu = \frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \delta \phi - \Lambda^\nu$$

despite this sign being incorrect, your final answer is right, due to the way you computed Λ^ν .

where Λ^ν is obtained by the direct substitution

$$\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \delta \phi = \partial_\nu \Lambda^\nu$$

$$\therefore \frac{\partial \mathcal{L}}{\partial \phi} a^\mu \partial_\mu \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi)} \partial_\nu \partial_\mu a^\mu \phi$$

$$= a^\mu \left(\frac{\partial \mathcal{L}}{\partial \phi} \partial_\mu \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi)} \partial_\nu \partial_\mu \phi \right) = a^\mu \delta_\mu^\nu \left(\frac{\partial \mathcal{L}}{\partial \phi} \partial_\nu \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi)} \partial_\nu \partial_\nu \phi \right)$$

Identifying the terms in brackets as the chain rule for multiple variables

$$= a^\mu \delta_\mu^\nu \partial_\nu \mathcal{L} = \partial_\nu (\delta_\mu^\nu a^\mu \mathcal{L}) \equiv \partial_\nu A^\nu$$

$\therefore A^\nu = \delta_\mu^\nu a^\mu \mathcal{L}$ or.... This procedure is kind of circular, the A^ν term is totally pinned down by the Noether's theorem, it should not be necessary to impose $\partial_\mu J^\mu = 0$ to get it. The answer is correct, but it shows a

Defining the energy-momentum tensor conceptual gap in the understanding of the Noether's theorem.

$$T_\mu^\nu := \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi)} \partial_\mu \phi - \delta_\mu^\nu \mathcal{L}$$

For our case

$$T_\mu^\nu = -F^{\nu\sigma} \partial_\mu A_\sigma + \delta_\mu^\nu \frac{F_{\rho\sigma} F^{\rho\sigma}}{4}$$

2/2

3. In order to see if the symmetry

$$\mathcal{L} \rightarrow \mathcal{L}' = \mathcal{L}(\phi', \partial^\mu \phi')$$

Note that $x \rightarrow e^{\epsilon} x \Rightarrow \partial_\mu \rightarrow \partial'_\mu = e^{-\epsilon} \partial_\mu$

$$\begin{aligned}\mathcal{L} &= (\partial'^\mu \phi' \partial'_\mu \phi') \frac{1}{2} - \frac{\lambda}{4!} (\phi')^4 \\ &= \frac{1}{2} (e^{-4\epsilon} \partial^\mu \phi \partial_\mu \phi) - \frac{\lambda}{4} e^{-4\epsilon} \phi^4 \\ &\approx e^{-4\epsilon} \mathcal{L}\end{aligned}$$

Furthermore, $S' = \int d^4x' \mathcal{L}' = \int d^4x e^{4\epsilon} \mathcal{L}' = \int d^4x e^{4\epsilon} e^{-4\epsilon} \mathcal{L} = S$;
 since the action S still the same, it is a symmetry. OK

To obtain the conserved current we invoke the Noether's Theorem. From the transformation of the coordinate

$$x^\mu \rightarrow x'^\mu = e^\epsilon x^\mu \approx (1 + \epsilon) x^\mu \therefore \delta x^\mu = \epsilon x^\mu$$

$$\begin{aligned}\Rightarrow \phi(x) &\rightarrow \phi'(x') = e^{-\epsilon} \phi(x) \approx (1 - \epsilon) \phi(x) \\ \phi'(x + \epsilon x) &\approx \phi'(x) + \epsilon x^\mu \partial_\mu \phi(x)\end{aligned}$$

$$\Rightarrow \phi'(x) + \epsilon x^\mu \partial_\mu \phi(x) = \phi(x) - \epsilon \phi(x)$$

$$\therefore \delta \phi = \phi'(x) - \phi(x) = -\epsilon (\phi(x) \ominus x^\mu \partial_\mu \phi(x))$$

wrong sign

To obtain Δ^ν

$$\delta S = \int d^4x' \mathcal{L}(\phi'(x'), \partial'_\mu \phi'(x')) - \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)$$

$$= \int d^4x e^{4\epsilon} \mathcal{L}(\phi'(x'), \partial'_\mu \phi'(x')) - \mathcal{L}(\phi, \partial_\mu \phi)$$

From the transformations $\Delta \phi = \phi'(x') - \phi(x) = -\epsilon \phi(x)$ OK

$$\approx \int d^4x (1 + 4\epsilon) \mathcal{L}(\phi + \Delta \phi, \partial_\mu \phi + \Delta(\partial_\mu \phi)) - \mathcal{L}(\phi, \partial_\mu \phi)$$

$$= \int d^4x (1 + 4\epsilon) \left(\mathcal{L}(\phi, \partial_\mu \phi) + \frac{\partial \mathcal{L}}{\partial \phi} \Delta \phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \Delta(\partial_\mu \phi) \right) - \mathcal{L}(\phi, \partial_\mu \phi)$$

Retaining the first order terms

$$\delta S = \int d^4x \left[4\epsilon \mathcal{L} + \frac{\partial \mathcal{L}}{\partial \phi} \Delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta (\partial_\mu \phi) \right]$$

Evaluating

$$\begin{aligned} \Delta (\partial_\mu \phi) &= \partial'_\mu \phi'(x') - \partial_\mu \phi(x) = e^{-\epsilon} \partial'_\mu \phi'(x') - \partial_\mu \phi(x) \\ &\stackrel{\sim}{=} (1-\epsilon) \partial_\mu [(1+\epsilon)\phi] - \partial_\mu \phi \\ &= (1-2\epsilon) \partial_\mu \phi - \partial_\mu \phi \\ &= -2\epsilon \partial_\mu \phi \end{aligned}$$

$$\therefore \delta S = \int d^4x \left[4\epsilon \mathcal{L} - \epsilon \frac{\partial \mathcal{L}}{\partial \phi} \phi - 2\epsilon \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi \right]$$

$$= \int d^4x \epsilon \left[4\mathcal{L} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \phi - 2 \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi \right]$$

$$= \int d^4x \epsilon \left\{ -\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \phi - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi + \left(4\mathcal{L} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi \right) \right\}$$

$$= \int d^4x \epsilon \left\{ -\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \phi + \left(4\mathcal{L} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi \right) \right\} = 0$$

Focusing on the second part; note that

$$\partial_\mu (x^\mu \mathcal{L}) = (\partial_\mu x^\mu) \mathcal{L} + x^\mu \partial_\mu \mathcal{L} = 4\mathcal{L} + x^\mu \partial_\mu \mathcal{L}$$

$$= 4\mathcal{L} + x^\mu \left(\frac{\partial \mathcal{L}}{\partial \phi} \partial_\mu \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\nu \partial_\mu \phi \right)$$

$$= 4\mathcal{L} + x^\mu \left(\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \right) \partial_\mu \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\nu (\partial_\mu \phi) \right)$$

$$= 4\mathcal{L} + x^\mu \left(\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi \right) \right)$$

$$\therefore 4\mathcal{L} = \partial_\mu (x^\mu \mathcal{L}) - x^\mu \left(\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi \right) \right) \sim$$

$$\delta S = \int d^4x \left\{ -\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \phi - x^\mu \mathcal{L} \right) - x^\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi \right) - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \phi \right\}$$

Now we may note that

$$\partial_\nu \left(x^\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi \right) = x^\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi \right) + \cancel{\partial_\nu x^\mu} \frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\mu \phi$$

Therefore

$$\delta S = \int d^4x \left\{ \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \phi - x^\mu \mathcal{L} + x^\nu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi \right) + \cancel{\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\nu \phi} - \cancel{\frac{\partial \mathcal{L}}{\partial (\partial_\nu \phi)} \partial_\nu \phi} \right\}$$

Since $\delta S = 0$ (is a symmetry)

$$\Rightarrow -\epsilon \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \phi - x^\mu \mathcal{L} + x^\nu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi \right) = 0 \quad \text{perfect!}$$

So then

here is correct

$\partial_\mu J^\mu = 0$ does NOT

NO !! imply $J^\mu = \text{cte}$

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \phi - x^\mu \mathcal{L} + x^\nu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi = \text{cte}$$

is the conserved current; or since $\mathcal{L}$ is known

$$J^\mu = (\partial^\mu \phi) \phi - x^\mu \mathcal{L} + x^\nu (\partial^\mu \phi) (\partial_\nu \phi)$$

$$= (\partial^\mu \phi) \phi - x^\mu \left(\frac{1}{2} (\partial^\nu \phi) (\partial_\nu \phi) - \frac{1}{4!} \phi^4 \right) + x^\nu (\partial^\mu \phi) (\partial_\nu \phi)$$

OK

4. A) For a Lorentz Transformation $X^\mu X_\mu = X'^\nu X'_\nu$;
 therefore for an infinitesimal transformation

1,8/2

$$X^\mu X_\mu = x'^\mu x'_\mu = g_{\alpha\beta} \Lambda^\alpha_\mu X^\mu \Lambda^\beta_\nu X^\nu$$

$$\stackrel{||}{=} x^\mu g_{\mu\nu} X^\nu$$

$$\therefore g_{\mu\nu} = g_{\alpha\beta} \Lambda^\alpha_\mu \Lambda^\beta_\nu$$

$$\Rightarrow g_{\mu\nu} = g_{\alpha\beta} (\delta^\alpha_\mu + \delta\omega^\alpha_\mu) (\delta^\beta_\nu + \delta\omega^\beta_\nu)$$

$$g_{\mu\nu} = g_{\mu\nu} + \delta\omega_{\nu\mu} + \delta\omega_{\mu\nu} + O(\delta\omega^2)$$

Retaining only the first order terms

$$\delta\omega_{\nu\mu} = -\delta\omega_{\mu\nu} \quad \text{OK}$$

B) The rotations are described by $\delta\omega_{ij}$; where $\delta\omega_{ii} = 0$ to $\delta\omega_{\mu\nu}$ be antisymmetric; for an infinitesimal rotation (spatial)

$$x'^i = \delta\omega^i_j x^j + x^i \Rightarrow \delta x^i = \delta\omega^i_j x^j$$

In the other hand; for a rotation $\delta x^i = (\delta\theta \hat{n}) \times x$
 $= \epsilon^i_{jk} \delta\theta^j x^k$

$$\therefore \delta\omega^i_k x^k = \epsilon^i_{jk} \delta\theta^j x^k$$

$$\delta\omega^i_k = \epsilon^i_{jk} \delta\theta^j \quad \text{OK}$$

c) For an infinitesimal transformation, the field is unchanged

$$\phi(x) \rightarrow \phi'(x') = \phi(x)$$

$$\Rightarrow \phi'(x + \delta x) = \phi(x)$$

$$\approx \phi'(x) + \partial_\mu \phi(x) \delta x^\mu = \phi(x)$$

$$\therefore \delta \phi = \phi'(x) - \phi(x) = -\partial_\mu \phi \delta x^\mu = (\partial_\mu \phi) \omega^{\mu\nu} x_\nu$$

correct sign!
look back at (2)

* For a rotation; $J = 1$

$$\delta S = \int d^4x [\mathcal{L}' - \mathcal{L}] = \int d^4x \partial_\nu A^\nu \rightarrow \text{This is what you should have done in (2)}$$

But $\mathcal{L}'(x) = \mathcal{L}(x + \delta x) = \mathcal{L} + \partial_\mu \mathcal{L} \delta x^\mu$
no, $\mathcal{L}' - \mathcal{L} = -\delta x^\mu \partial_\mu \mathcal{L}$

$$\Rightarrow \mathcal{L}' - \mathcal{L} = (\partial_\mu \mathcal{L}) \omega^{\mu\nu} x_\nu$$

$$\delta S = \int d^4x' \mathcal{L}'(x') - \int d^4x \mathcal{L}(x) = -\omega^{\mu\nu} \partial_\mu (\mathcal{L} x_\nu) + \omega^{\mu\nu} \mathcal{L} (\partial_\mu x_\nu)$$

$$= \int d^4x \mathcal{L}'(x) - \int d^4x \mathcal{L}(x) = -\omega^{\mu\nu} \partial_\mu (\mathcal{L} x_\nu) + \omega^{\mu\nu} \mathcal{L} (\partial_\mu x_\nu)$$

$$= \int d^4x (\mathcal{L}'(x) - \mathcal{L}(x)) = \partial_\mu (\omega^{\mu\nu} \mathcal{L} x_\nu) \quad \text{is Antisymmetric}$$

$$\Lambda^\mu = -\omega^{\mu\nu} x_\nu \mathcal{L}$$

The Noether current is then

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi - \Lambda^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} (-\omega^{\nu\beta} x_\beta \partial_\nu \phi) - \omega^{\mu\beta} x_\beta \mathcal{L}$$

$$= -\omega^{\nu\beta} x_\beta \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} \right] = -\omega^{\nu\beta} x_\beta \Theta_\nu^\mu$$

Energy-Momentum Tensor appears if you write correctly at

Since $\omega^{\nu\beta}$ is antisymmetric $\Lambda^\mu = -\omega^{\mu\nu} x_\nu \mathcal{L}$

$$J^\mu = \frac{-\omega^{\nu\beta}}{2 \omega^{\text{arbitrary}}} [x_\beta \Theta_\nu^\mu - x_\nu \Theta_\beta^\mu]$$

$$\therefore \mathcal{M}_{\beta\nu}^\mu := x_\beta \Theta_\nu^\mu - x_\nu \Theta_\beta^\mu \quad \text{OK}$$

5. For this situation
1,8/2

$$A^\mu(x') \approx (\delta_\nu^\mu + \omega_\nu^\mu) A^\nu(x)$$

$$\Rightarrow A^\mu(x + \delta x) \approx A^\mu(x) + \omega_\nu^\mu A^\nu(x)$$

$$\Rightarrow A^\mu(x) + \partial_\rho A^\mu(x) \delta x^\rho = A^\mu(x) + \omega_\nu^\mu A^\nu$$

$$\therefore A^\mu(x) - A^\mu(x) = \delta A^\mu = -(\partial_\rho A^\mu) \omega_\nu^\rho x^\nu + \omega_\nu^\mu A^\nu$$

Therefore the new Noether's current

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu)} \delta A^\nu$$

wrong, it is +, some problem in the last exercise.

a sign disappeared...

$$= \frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu)} [(\partial_\rho A^\nu) \omega_\sigma^\rho x^\sigma + \omega_\nu^\rho A^\rho] - \omega_\nu^\mu x^\nu \mathcal{L}$$

$$= \omega^{\rho\sigma} x_\sigma \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu)} \partial_\rho A^\nu - \delta_\rho^\mu \mathcal{L} \right] + \omega^{\nu\rho} \frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu)} A_\rho$$

final result is correct due to you changing two signs

The first term is the same as in the previous exercise

$$J^\mu = \omega^{\rho\sigma} [x_\sigma \otimes_\rho^\mu] + \omega^{\nu\rho} \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\nu)} A_\rho \right]$$

Using the antisymmetry expansion

$$J^\mu = \underbrace{\omega^{\rho\sigma}}_{\substack{2 \\ \text{arbitrary}}} [x_\sigma \otimes_\rho^\mu - x_\rho \otimes_\sigma^\mu + \underbrace{\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\rho)} A_\sigma - \frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\sigma)} A_\rho}_{S^\mu_{\sigma\rho}}]$$

$$\therefore M^\mu_{\rho\sigma} := [x_\sigma \otimes_\rho^\mu - x_\rho \otimes_\sigma^\mu + S^\mu_{\sigma\rho}] \text{ ok, usually is defined with opposite sign}$$

For our case, using previous results $\frac{\partial \mathcal{L}}{\partial(\partial_\mu A^\nu)} = -F^\mu_\nu$

please distinguish, $F^\mu_\nu \neq F_\nu^\mu$

$$\therefore \textcircled{H} \quad \underline{\mathcal{M}}^\mu_\rho = -F^\mu_\nu \partial_\rho A^\nu + \frac{\delta^\mu_\rho}{4} F^{\alpha\beta} F_{\alpha\beta} \quad \rightarrow \text{wrong}$$

$$S^\mu_{\sigma\rho} = -F^\mu_\rho A_\sigma + F^\mu_\sigma A_\rho$$

$\Rightarrow$

$$\begin{aligned} \mathcal{M}^\mu_{\sigma\rho} &= x_\rho F^\mu_\nu \partial_\sigma A^\nu + \frac{\delta^\mu_\sigma}{4} F^{\alpha\beta} F_{\alpha\beta} x_\rho - x_\sigma F^\mu_\nu \partial_\rho A^\nu - \frac{\delta^\mu_\rho}{4} F^{\alpha\beta} F_{\alpha\beta} x_\sigma \\ &\quad + F^\mu_\sigma A_\rho - F^\mu_\rho A_\sigma \quad \mathcal{N} \end{aligned}$$

$$\begin{aligned} \mathcal{M}^\mu_{\sigma\rho} &= F^\mu_\nu (x_\rho \partial_\sigma A^\nu - x_\sigma \partial_\rho A^\nu + \delta^\nu_\sigma A_\rho - \delta^\nu_\rho A_\sigma) + \\ &\quad + \frac{F^{\alpha\beta} F_{\alpha\beta}}{4} (\delta^\mu_\sigma x_\rho - \delta^\mu_\rho x_\sigma) \quad \mathcal{N} \end{aligned}$$